



Scenario Based Design

Deep Dive 1



Design For Complex system

- Addressing ill-defined problems
- Possible move are not defined their consequences are undefined
- To understand the problem designer/engineers will tend to focus on solution first and will encounter functional fixation
- The problem tend to be reduced to the scope of the solution or the technology
- "Getting the right design before getting the design right" - Tohidi

Scenario based Design

- Scenario-based design is a family of techniques in which the **use** of a future system is concretely described at an early point in the development process



Scenario: Car Appointment

- Dimitri a teacher at Estia for 5 years must visit Antoine an apprentice working in Pau. He creates an appointment in his calendar and invites the student and the company car to reserve it. From the student information, the event is updated to consider the travel time. On the day of the visit, Dimitri receives a GPS location which allows him to find the company car. The car is unlocked thanks to his phone. The traffic is normal, the GPS of the car tells him to join the highway in the direction of Pau. Before the separation of the highway, the GPS reminds him not to take the road that he usually take to get home but to exit on the A64. Near Pau the GPS goes back in step-by-step mode to guide Dimitri to its destination. On the way back the GPS plans a trajectory through a partner gas station. The final mileage and the position of the car are automatically updated in the event.

Scenarios

- Scenarios are short and concise stories
 - They consist of a declarative part describing the **initial situation, context or setting**
 - **Actors** with **goals** and capabilities
 - Physical resources
 - A procedural part describing a sequence of actions, tasks and events that leads to a final situation
- **Scenarios are :**
 - Easy to produce and communicate
 - Concrete And Flexible, Tangible and vague
 - Evocative, enabling reflection
 - Centered on use, hence on human
 - Heuristics
 - Scenarios can be used at many levels of details, of regarding different perspectives.

AS-IS Scenarios

- Problem scenarios explicit the context, goals, agents and their capabilities.
 - Synthesis and explicit knowledge from the analysis phase (ethnographic, interviews, desk research, state of the art)
- **Front desk reservation**
 - Dimitri contacts the student to arrange the meeting, Antoine tells him he work at Pau, they agree on a meeting in the afternoon to accommodate for the travel. Dimitri contacts Sylvie at the reception to book a car for the day. On the morning of the event, he gets the key from the front desk as well as the carnet-de-bord where the car location is written very broadly. In the car he take some time to find the address in his mail and set up the GPS, he puts the GPS on mute as it giving to much information. On the way back, he leaves the highway one exit further to reach the only partner gas station he knows and double back. Once parked at estia, he copy the previous end mileage as his start and writes the current mileage as his end mileage. It is late he drops the keys and carnet de bord in the reception mailbox

TO-BE SCENARIOS

- Explicit the use of the system to be designed.
- Several level of detail or perspectives :
 - Activity/Task – what are the goals, tasks/subtask
 - Information – How the information is accessible to the user
 - Interaction – Exchanges between users and the systems
- Car Appointment
 - Dimitri a teacher at Estia for 5 years must visit Antoine an apprentice working in Pau. He creates an appointment in his calendar and invites the student and the company car to reserve it. From the student information, the event is updated to consider the travel time. On the day of the visit, Dimitri receives a GPS location which allows him to find the company car. The car is unlocked thanks to his phone. The traffic is normal, the GPS of the car tells him to join the highway in the direction of Pau. Before the separation of the highway, the GPS reminds him not to take the road that he usually take to get home but to exit on the A64. Near Pau the GPS goes back in step-by-step mode to guide Dimitri to its destination. On the way back the GPS plans a trajectory through a partner gas station. The final mileage and the position of the car are automatically updated in the event.

Error Scenario

- Make explicit what could go wrong
- Describe anormal situation or even critical/urgent situation
- Low Battery:
- <Dimitri situation and goals>
- <Initial rendez-vous procedure and car pick up>
- While Driving in GPS mode the battery of the phone reach critically low level. The phone instructs the car of the final destination and indicate the user to exit at the next highway rest area to plug the phone.

CLAIM ANALYSIS

Analysing scenarios to elicit consequences, advantages or issues

Considers both positive and negative consequences

Allows iteration process to define scenario and solution that alleviate the negative consequences and enhance the positive consequences

Going through the reception desk to book a car

- + Can rent a car if no car is available
- reception must be open (9am-5pm)

Booking the car through the calendar

- + Direct interaction, easy to reschedule a visit according to availability of the car
- No conflict management or priority booking
- Not possible to

Formalisms

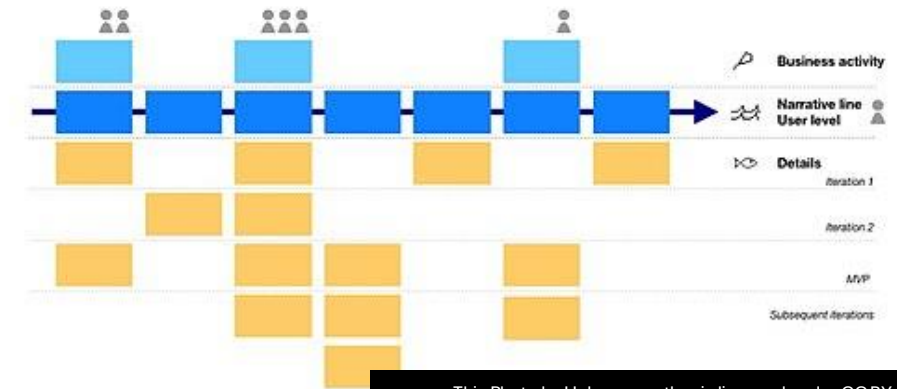
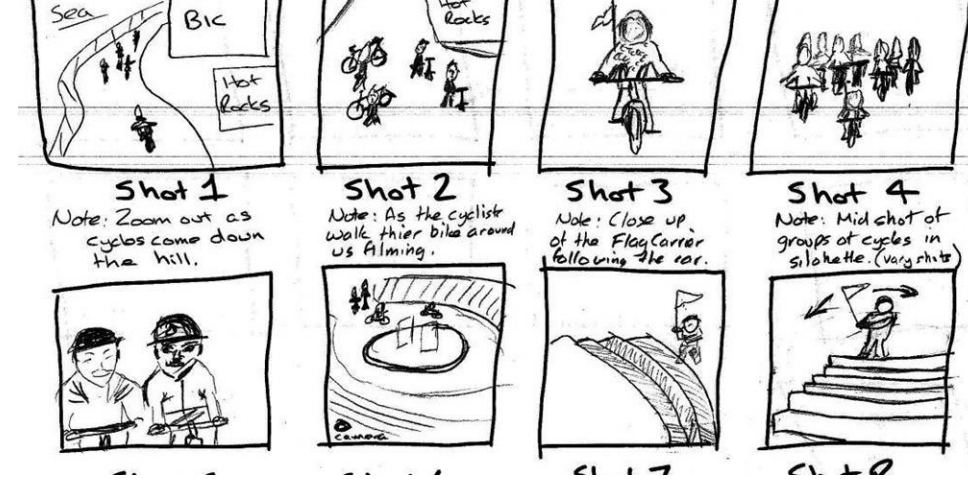
Free form text

Storyboard

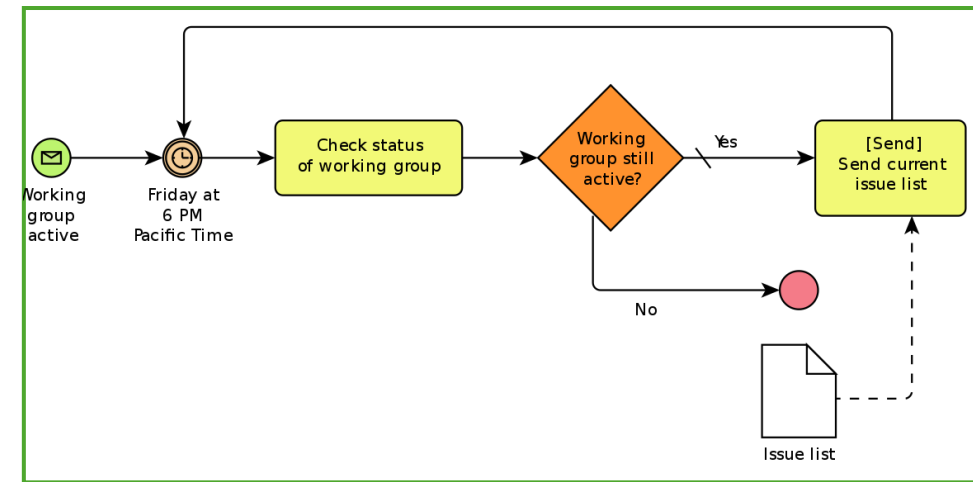
User stories

BPMN

UML – activity diagrams / sequence diagrams



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SCENARIO BUILDING Methods



ETHNOGRAPHIC FIELD STUDY

- Going in the "field" to observe users
- Build ontologies of actors, goals, event from scratch
- Can be extended with Think aloud, interviews, focus group
- Discover failure mode, works around, design opportunities
- Involve immersion, can take time



PARTICIPATORY DESIGN

- Involve users as *designers* not just to evaluate prototypes
- Complementary with ethnographic studies
- Useful to elicit explicit needs
- Improve adhesion with solutions
- But participants are not actual designer/engineer. Deeper analysis is required
- See also: creative sessions in particular GEM sessions, Improv prototyping...



Reusing Materials



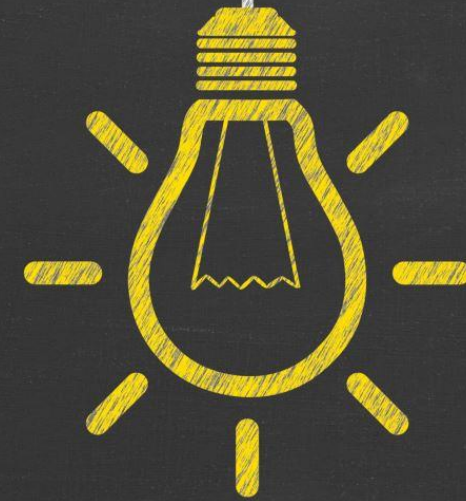
- Using prior analysis
- Using recorded procedures or formations, tutorials etc.
- Working by analogies with other domains
- Build upon use of other tools / systems – Redesign
- Build typologies to ease later reuse of scenario

THEORY BASED

- Based on Activity theory, sociology or psychology theories
- Design theories : heuristic, prospective ergonomic, human factors
- See. Cognitive Engineering Class (ex. Cognitive Function Analysis)
- Build typology or domain ontology from scenario

Creative SESSION

- Brainstorming session
- Improv prototyping
- transforming scenarios :
 - by asking what if ?
 - What could go wrong ?
 - Changing the point of view
 - Changing the resources (physical or cognitive)
- Analogy : in our case study the car as a person, the car as a resources etc



TECHNOLOGY BASED

- New technologies creates new affordances (See introduction class)
- Revolution are hard to identify in ethnographic studies or participatory design
- State of the art review
- In the end it must be validated against use

